MARKING SCHEME

Model Paper — 03

Mathematics I

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Note (for teachers):

- Part A: 2 marks for each question (usually 1 mark for a correct method and 1 mark for the correct final answer). Total 50 marks.
- Part B: 10 marks for each question. Total 50 marks. The marks for each step are shown in the right-hand column; the step marks add up to the marks printed for each sub-part.
- Give full marks for correct alternative methods. For values read from graphs or measured from constructions, accept a reasonable range.

Part A — Answers

1. $12 \times 15 = n \times 10 \Rightarrow n = 18$ workers; extra $= 18 - 12$

Final answer: 6 workers

(2 marks)

2. Through E draw a line parallel to AB (and CD): $\angle BAE + \angle ECD = \angle AEC \Rightarrow \angle BAE = 115° - 45°$

Final answer: $\angle BAE = 70°$

(2 marks)

3. $\frac{3\times3 + 5\times2 - 2\times4}{12x} = \frac{9 + 10 - 8}{12x}$

Final answer: $\frac{11}{12x}$

(2 marks)

4. The faces of a right triangular prism are two triangles and rectangles. A parallelogram with no right angle is not a rectangle.

Final answer: (iii) ticked

(2 marks)

5. 8 students $\leftrightarrow 20° \Rightarrow 1° \leftrightarrow 0.4$ students; athletics: $180° \Rightarrow 180 \times 0.4$ (or 8×9)

Final answer: 72 students

(2 marks)

6. $\angle ADB = \frac{1}{2} \angle AOB$; $\angle AEB + \angle ADB = 180°$ (cyclic) $\Rightarrow \angle ADB = 40°$; CD bisects $\angle ADB$ (symmetry, AC = CB)

Final answer: $\angle CDB = 20°$

(2 marks)

7. $3(x^2 - 25) = 3(x - 5)(x + 5)$

Final answer: $3(x - 5)(x + 5)$

(2 marks)

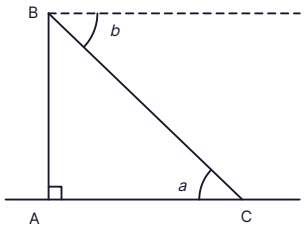
8. $\lg 6000 = \lg 6 + \lg 1000 = 0.7782 + 3$

Final answer: 3.7782

(2 marks)

9. a marked at C (between CA and CB) and b marked at B (between the dashed horizontal and BC); alternate angles

Final answer: $a = b$



(2 marks)

10. Curved surface = $2\pi r(3r) = 6\pi r^2$; base = πr^2; ratio = $6\pi r^2 : \pi r^2$

Final answer: 6 : 1

(2 marks)

11. Total outcomes = $4 \times 4 = 16$; sum 10: (3, 7), (5, 5), (7, 3) = 3 outcomes

Final answer: $\frac{3}{16}$

(2 marks)

12. Exterior angle: $\angle DBC = 40° + 40° = 80°$ (AB = BD); DB = DC $\Rightarrow$ $\angle DCB = 80°$; $x = 180° - 80° - 80°$

Final answer: $x = 20°$

(2 marks)

13. $A \cap B = \{6\}$ ($A = \{3, 6, 9\}$, $B = \{2, 4, 6, 8, 10\}$); $A \cup B = \{2, 3, 4, 6, 8, 9, 10\}$, n = 7

Final answer: (ii) ticked

(2 marks)

14. $\frac{24x^6y^4}{8xy^2}$

Final answer: $3x^5y^2$

(2 marks)

15. Common ratio $r = (-12) \div (-4) = 3$; $T_4 = -36$, $T_5 = -108$ (or $T_5 = T_3 \cdot r^2 = -12 \times 9$)

Final answer: −108

(2 marks)

16. ∠BDC = ∠BAC = 28° (same segment); ∠ADC = 2 × 28° = 56°; ∠CBE = ∠ADC (exterior angle of a cyclic quadrilateral)

Final answer: ∠CBE = 56°

(2 marks)

17. Coefficients: HCF(8, 12, 20) = 4; a: lowest power a^1; b: lowest power b^1; c is not common

Final answer: 4ab

(2 marks)

18. Prism volume = 38.5 × h = $\pi r^2 h \Rightarrow (22/7)\, r^2 = 38.5 \Rightarrow r^2 = 38.5 \times 7/22 = 12.25$

Final answer: r = 3.5 cm

(2 marks)

19. $DE^2 = 10^2 - 6^2 = 64 \Rightarrow$ DE = 8 cm; BC = 2DE = 16 cm; BD = 6 cm; CE = AE = 10 cm. Perimeter = 16 + 10 + 8 + 6

Final answer: 40 cm

(2 marks)

20. Gradient = (13 − 4)/(5 − 2) = 3; y − 4 = 3(x − 2)

Final answer: y = 3x − 2

(2 marks)

21. ∠DAB = ∠BCD = 64° (opposite angles of a parallelogram); ∠AEB = 90° (angle in a semicircle); ∠ABE = 180° − 90° − 64°

Final answer: ∠ABE = 26°

(2 marks)

22. Amount at end of year 1 = 21 600 − 3 600 = Rs 18 000; 18 000 = P × (120/100) ⇒ P = 18 000 × 100/120

Final answer: Rs 15 000

(2 marks)

23. ∠OBD = ∠AOB = 70° (alternate angles, OA ∥ BD); OB = OD ⇒ ∠ODB = 70°, ∠BOD = 180° − 140° = 40°; ∠BED = ½ ∠BOD

Final answer: ∠BED = 20°

(2 marks)

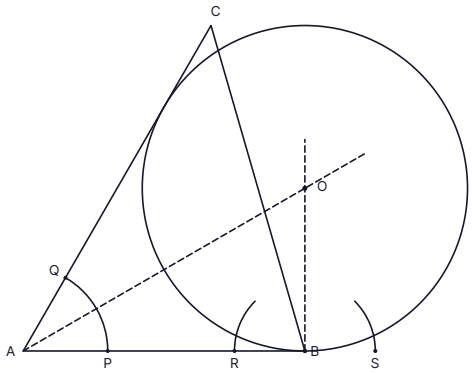
24. $5x > 3 - 28 = -25 \Rightarrow x > -5$

Final answer: −4, −3, −2, −1

(2 marks)

25. Bisect ∠BAC (arcs from P and Q meet; draw the bisector from A); draw the perpendicular to AB at B (arcs from R and S meet; draw the line through B). They meet at O.

Final answer: O is the intersection; OB ≈ 3.5 cm (3.3 – 3.7 accepted)



(2 marks)

Part B — Marking scheme

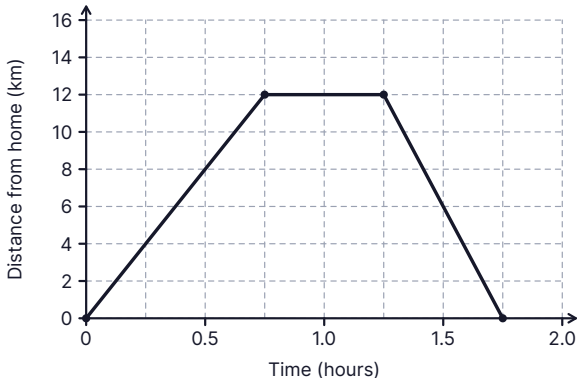
Question B1

(i)	
Remainder after food = $1 - \frac{1}{5} = \frac{4}{5}$	**1**
(ii)	
Spent on other expenses = $\frac{3}{8}$ of $\frac{4}{5} = \frac{3}{10}$ of the money received (method)	**1**
Remaining = $\frac{4}{5} - \frac{3}{10} = \frac{1}{2}$ (or $\frac{4}{5} \times \frac{5}{8} = \frac{1}{2}$)	**1**
(iii)	
Money received = 95 000 × 2 = Rs 190 000; food = $\frac{1}{5}$ × 190 000 = **Rs 38 000**	**1**
Remainder after food = 190 000 − 38 000 = 152 000; other expenses = $\frac{3}{8}$ × 152 000 = **Rs 57 000**	**1**
(iv)	
Let the part of the salary above Rs 100 000 be Rs y. Tax = 10% of y. Salary − tax = money received: $100\,000 + y - 0.1y = 190\,000$	**1**
$0.9y = 90\,000 \Rightarrow y = 100\,000$	**1**
Salary = 100 000 + 100 000 = **Rs 200 000** (tax paid Rs 10 000 — check: 200 000 − 10 000 = 190 000)	**1**
(v)	
$\frac{38\,000}{200\,000} \times 100$	**1**
19%	**1**

Total marks: 10

Question B2

(a)(i)	
Horizontal line from (0.75, 12) to (1.25, 12) (staying 30 minutes)	**1**
Return segment ends at the correct time 1.25 + 0.5 = 1.75 h, at distance 0 (point (1.75, 0))	**1**
Straight line segments joined correctly: (0.75, 12) → (1.25, 12) → (1.75, 0)	**1**
(a)(ii)	
Speed = distance ÷ time = 12 ÷ 0.5 (or $\frac{12}{30/60}$)	**1**
24 km/h	**1**
(b)(i)	
Length of the semicircular arc = π × 3.5 = $\frac{22}{7}$ × 3.5 (or ½ × 2πr)	**1**
= 11 cm	**1**
Perimeter = 11 + 22 + 7 + 22 = **62 cm** (AB is not part of the boundary)	**1**
(b)(ii)	
Area of rectangle = 7 × 22 = 154 cm²; area of semicircle = ½ × $\frac{22}{7}$ × 3.5 × 3.5 = 19.25 cm²	**1**
154 ÷ 19.25 = **8 times**	**1**



Correct graph (above): (0, 0) → (0.75, 12) → (1.25, 12) → (1.75, 0). Accept points within half a small square.

Total marks: 10

Question B3

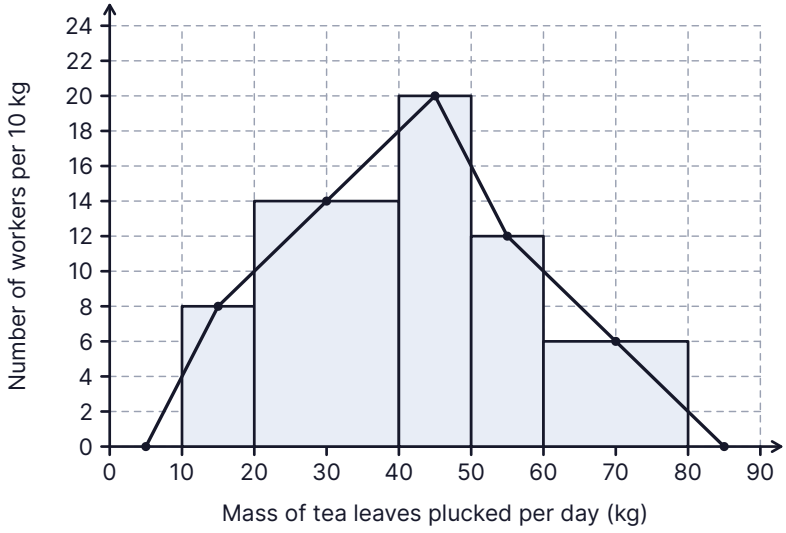
(i)	
Rates = $\frac{20}{100}$ × 88 000 (on the assessed annual value)	**1**
= **Rs 17 600**	**1**
(ii)	
Annual rent = 9 500 × 12 = Rs 114 000; maintenance = 10% × 114 000 = Rs 11 400	**1**
Remaining = 114 000 − 17 600 − 11 400 = **Rs 85 000**	**1**
(iii)(a)	
Number of shares = $\frac{7\,000}{3.50}$	**1**
= **2 000 shares**	**1**
Amount invested = 2 000 × 50 = **Rs 100 000**	**1**
(iii)(b)	
Extra money added = 100 000 − 85 000 = Rs 15 000	**1**
$\frac{15\,000}{100\,000}$ × 100	**1**
15%	**1**

Total marks: 10

Question B4

(i)	
Number = height × (class width ÷ 10): 20 – 40: 14 × 2 = **28**	**1**
40 – 50: **20** and 50 – 60: **12** (width 10, so height = number)	**1**
60 – 80: 6 × 2 = **12** (total 8 + 28 + 20 + 12 + 12 = 80 ✓)	**1**
(ii)	
Five rectangles with the correct class boundaries 10, 20, 40, 50, 60, 80 on the horizontal axis, joined side by side (no gaps)	**1**
Heights 8, 14, 20, 12, 6 — all five correct (any three or four correct: 1 mark)	**2**
(iii)	
Cumulative frequencies: 8, 36, 56 (first three correct)	**1**
68, 80 (last two correct)	**1**
(iv)	
Workers with 50 kg or more = 12 + 12 = 24	**1**
Probability = $\frac{24}{80} = \mathbf{\frac{3}{10}}$	**1**

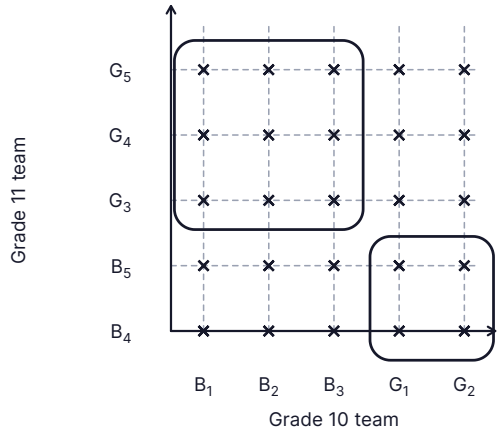
Completed table (class | number | cumulative): 10 – 20 | 8 | 8; 20 – 40 | 28 | 36; 40 – 50 | 20 | 56; 50 – 60 | 12 | 68; 60 – 80 | 12 | 80.

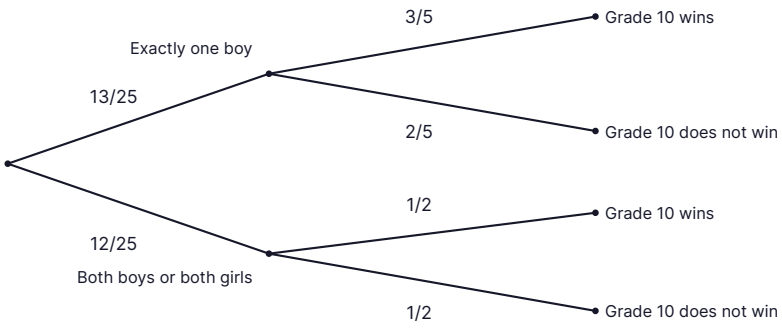


Total marks: 10

Question B5

(i)	
All 25 intersection points marked with "×" (23 – 24 correct: 1 mark)	**2**
(ii)	
Closed curves enclosing exactly the 13 outcomes: the block B_1, B_2, B_3 × G_3, G_4, G_5 (9 outcomes) and the block G_1, G_2 × B_4, B_5 (4 outcomes)	**1**
Favourable outcomes 9 + 4 = 13; all outcomes 25	**1**
P(exactly one boy) = $\mathbf{\frac{13}{25}}$	**1**
(iii)	
First branches: $\frac{13}{25}$ (exactly one boy) and $\frac{12}{25}$ (both boys or both girls)	**1**
After "exactly one boy": wins $\frac{3}{5}$, does not win $\frac{2}{5}$	**1**
After "both boys or both girls": wins $\frac{1}{2}$, does not win $\frac{1}{2}$	**1**
(iv)	
P(does not win) = $\frac{13}{25} \times \frac{2}{5} + \frac{12}{25} \times \frac{1}{2} = \frac{26}{125} + \frac{6}{25}$	**1**
= $\mathbf{\frac{56}{125}}$	**1**





Total marks: 10